Chapter 7

Introduction to categories

7.1. The definition of a category

We have now seen many examples of representation theories and of operations with representations (direct sum, tensor product, induction, restriction, reflection functors, etc.). A context in which one can systematically talk about this is provided by **category theory**.

Category theory was founded by Saunders Mac Lane and Samuel Eilenberg in 1942—1945. It is a fairly abstract theory which seemingly has no content, for which reason it was christened "abstract nonsense" (see Section 7.10). Nevertheless, it is a very flexible and powerful language, which has become totally indispensable in many areas of mathematics, such as algebraic geometry, topology, representation theory, and many others.

We will now give a very short introduction to category theory, highlighting its relevance to the topics in representation theory we have discussed. For a serious acquaintance with category theory, the reader may use, for instance, the classical book [**McL**].

Definition 7.1.1. A **category** $\mathcal{C}$ is the following data:

(i) A class of objects $Ob(\mathcal{C})$.